COHOMOLOGY ON THE PUNCTURED SPECTRUM IS THE COLIMIT OF DUAL KOSZUL HOM GROUPS: THE COMPARISON MAP IS AN ISOMORPHISM

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Editorial note. Inputs lemma-affine-compare-bounded, lemma-alternating-cech-complex-complex and lemma-alternating-cech-complex-complex-computes-cohomology are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Allowed standard prerequisites (editorial).

- Affine comparison of D_QCoh with module complexes. lemma-affine-compare-bounded (same chapter, NOT carded, declared external). Represents E by a complex of quasi-coherent sheaves F-bullet with M-bullet = Gamma(X, F-bullet).
- Cech complex of the cover computes over alternated totals. lemma-alternating-cech-complex-complex and lemma-alternating-cech-complex-complex-computes-cohomology (same chapter, NOT carded, declared external). Give the two quasi-isomorphisms from the Hom-complex colimit to the alternated Cech total complex and to R Gamma(U, F-bullet).
- Hom out of the dual Koszul resolution in D(A). More on Algebra, lemma RHom-out-of-projective and equation h0-RHom (cross-chapter, named). H^0 of the Hom-complex computes Hom in D(A).

Source reference. Original equation 08DC: sources/stacks-perfect/pinned-original-source.tex, source lines 2308–2308. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08D1: sources/stacks-perfect/pinned-original-source.tex, source lines 2231–2231. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08D2: sources/stacks-perfect/pinned-original-source.tex, source lines 2272–2272. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A64: focused-seeds/geometry/S9-perf2-represent-cohomology-class-on-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A66: focused-seeds/geometry/S9-perf2-represent-cohomology-class-on-open/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08CZ: sources/stacks-perfect/pinned-original-source.tex, source lines 2116–2116. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *In Situation 08CZ. For every object E of $D_{QCoh}(\mathcal{O}_X)$ the map (08DC) is an isomorphism.*

Proof. By Lemma 06Z0 we may assume that E is given by a complex of quasi-coherent sheaves $\mathcal{F}^\bullet$. Let $M^\bullet = \Gamma(X, \mathcal{F}^\bullet)$ be the corresponding complex of A-modules. By Lemmas 08D1 and 08D2 we have quasi-isomorphisms

$$\operatorname{colim}_e \operatorname{Hom}^\bullet(I^\bullet(f_1^e, \ldots, f_r^e), M^\bullet) \longrightarrow \operatorname{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet)) \longrightarrow R\Gamma(U, \mathcal{F}^\bullet)$$

By More on Algebra, Lemma 0A66 and Equation (0A64) taking H^0 of the complex $\operatorname{Hom}^\bullet(I^\bullet(f_1^e, \ldots, f_r^e), M^\bullet)$ computes Hom in $D(A)$. Thus taking H^0 on both sides we obtain

$$\operatorname{colim}_e \operatorname{Hom}_{D(A)}(I^\bullet(f_1^e, \ldots, f_r^e), M^\bullet) = H^0(U, E)$$

Since $\operatorname{Hom}_{D(A)}(I^\bullet(f_1^e, \ldots, f_r^e), M^\bullet) = \operatorname{Hom}_{D(\mathcal{O}_X)}(I_e, E)$ by Lemma 06Z0 the lemma follows. □